✅ Assignment-1 E-commerce

Aim

To create E-commerce order program in Java.

Theory

This program uses class, object, constructor, and Scanner.
It calculates total bill and discount.

Program Logic

Take product details → calculate total → apply discount → display bill.

Syntax Used

- class
- object
- constructor
- this
- if
- Scanner

Algorithm

1. Input product details
2. Calculate total
3. Apply discount
4. Display output

Conclusion

Thus, E-commerce program was executed successfully.

✅ Assignment-2 Method Overloading

Aim

To implement method overloading in Java.

Theory

This program uses overloaded methods for power and absolute value.
Math class functions are also used.

Program Logic

Call different methods with different data types.

Syntax Used

- static
- method overloading
- Math.pow()
- Math.abs()

Algorithm

1. Create methods
2. Call methods
3. Display result

Conclusion

Thus, method overloading program was executed successfully.

✅ Assignment-3 Library Management

Aim

To create Library Management System in Java.

Theory

This program manages books using static variable and methods.

Program Logic

Add books → issue/return books → display total books.

Syntax Used

- static
- switch
- do-while
- array of objects

Algorithm

1. Add books
2. Display books
3. Issue/return books
4. Show total books

Conclusion

Thus, Library Management program was executed successfully.

✅ Assignment-4 Array Operations

Aim

To perform array operations in Java.

Theory

Array stores multiple values.
Program finds max, min, sum, average, and search element.

Program Logic

Input array → perform operations → display result.

Syntax Used

- array
- for loop
- if
- Scanner

Algorithm

1. Input array
2. Find max/min
3. Find sum/average
4. Search element

Conclusion

Thus, array operations program was executed successfully.

✅ Assignment-5 Single Inheritance

Aim

To implement single inheritance in Java.

Theory

Inheritance allows child class to use parent class methods.

Program Logic

Child class accesses parent class methods.

Syntax Used

- extends
- class
- object

Algorithm

1. Create parent class
2. Create child class
3. Call methods

Conclusion

Thus, inheritance program was executed successfully.

✅ Assignment-6 Interface

Aim

To implement interface and polymorphism.

Theory

Interface contains abstract methods.
Class implements interface methods.

Program Logic

Create interface → implement methods → call methods.

Syntax Used

- interface

- implements
- polymorphism

Algorithm

1. Create interface
2. Implement interface
3. Call methods

Conclusion

Thus, interface program was executed successfully.

✅ Assignment-7 Abstract Class

Aim

To implement abstract class in Java.

Theory

Abstract class contains abstract methods.
Child class provides implementation.

Program Logic

Create abstract class → implement method in child class.

Syntax Used

- abstract class
- abstract method
- inheritance

Algorithm

1. Create abstract class
2. Create child class
3. Call methods

Conclusion

Thus, abstract class program was executed successfully.

✅ Assignment-8 ATM Exception Handling

Aim

To implement ATM system using exception handling.

Theory

This program performs deposit, withdraw, and balance operations using try-catch.

Program Logic

Take user choice → perform transaction → handle exception.

Syntax Used

- try
- catch
- finally
- throw

Algorithm

1. Display menu
2. Take input
3. Perform operation
4. Handle exception

Conclusion

Thus, ATM program was executed successfully using exception handling.